

PARTS LIST

Semiconductors:

- IC1 - TL072 dual op-amp
 IC2, IC3 - LM1875 power amplifier
 ZD1, ZD2 - 12V, 1W zener diode
 BR1 - W04, 1.5A bridge rectifier module
 LED1 - 5mm light emitting diode

Resistors (all 1/4-watt, 5% carbon unless stated otherwise):

- R1, R5, R7, R15, R13 - 2.7-kilo-ohm
 R2, R16 - 100-ohm
 R3, R4, R17, R18 - 270-kilo-ohm
 R6, R14 - 5.1-kilo-ohm
 R8, R19, R10, R21 - 82-kilo-ohm
 R9, R20 - 4.7-kilo-ohm
 R11, R22 - 2.7-ohm
 R12, R23 - 470-ohm
 VR1, VR2 - 100-kilo-ohm log potentiometer

Capacitors:

- C1, C12 - 470pF ceramic
 C2, C13 - 100µF, 50V electrolytic
 C3, C5, C10, C16, C19, C21 - 0.1µF ceramic
 C4, C20 - 1000µF, 50V electrolytic
 C7, C14 - 1µF, 25V electrolytic
 C8, C15 - 4.7µF, 50V electrolytic
 C6, C17, C11, C22 - 220µF, 50V electrolytic
 C9, C18 - 0.22µF ceramic
 C23-C26 - 2200µF, 50V electrolytic

Miscellaneous:

- S1 - DPDT switch
 S2 - On/off switch
 F1 - 2A power fuse
 X1 - 230V AC primary to 21V-0-21V, 2A secondary centre-tapped transformer
 - Metal case
 - Finned heatsink
 - RCA connectors

is not a problem, but it is not the case with add-on amplifier for a stereo or bridged mono system.

The circuit described in the article provides two 15W RMS power amplifier channels, with low distortion noise and a good frequency response. The input sensitivity is good, too. You need around 200mV input to produce full output, suitable for use with 'line' output on DVD players with in-built Dolby Digital decoders.

In fact, one or two brick amplifiers can be used with the DVD player to get an effective output from your surround sound system. This amplifier is

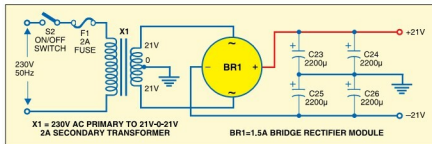


Fig. 2: Power supply circuit of stereo brick amplifier

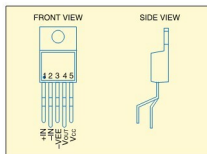


Fig. 3: Pin details of LM1875

also suitable for enhancing multimedia in a computer, allowing you to play external speakers.

How it works?

The brick amplifier circuit is simple. It consists of power supply and audio power amplifier sections. The simple scheme of the two channels to be driven from one of the inputs facilitates a hassle-free bridge mono mode operation.

The circuit of brick amplifier is shown in Fig. 1. It consists of a TL072 low-noise JFET dual operational amplifier (IC1) and two LM1875 (IC2 and IC3) power amplifiers.

In stereo mode, each half of the TL072 operates as a non-inverting input amplifier, with a gain of about 2.8 times as determined by feedback resistors R5 and R6. In front of each input amplifier there is a low-pass filter for RF suppression (R1/C1 and R15/C12), followed by volume controls VR1 and VR2 and blocking capacitors C2 and C13.

Switch S1 is used to select bridge mono mode operation. When it is switched to mono position 'M', the non-inverting input of IC1(B) is connected to the ground, making the left

input of the brick ineffective. However, at inverting terminal of IC1(B), the cold end of resistor R13 is disconnected from the ground and connected to R7. It is now converted into a unity gain inverting stage and the output of IC1(A) is connected to IC1(B) through R7. This converts the front end to produce dual phase drive signals from the right channel input. In stereo mode, each of the amplifier stage drives the right and the left speaker connected across pin 4 of IC2 and IC3, respectively.

Each input stage drives one of the LM1875 power stages (IC2 and IC3) with traditional circuitry. Coupling capacitors C7 and C14 provide DC blocking, with the power stage voltage gains set to around 18 by feedback networks R9/R10 and R20/R21. Resistors R8 and R19 provide biasing for the non-inverting inputs, while C8/C15 provide AC grounding for the feedback networks.

R11 and C9, and R22 and C18, are Zobel impedance stabilising circuits used across power amplifier output to ensure maximum AC stability.

Output stages operate from DC rails of $\pm 21V$, which are provided from the power supply, using 21V-0-21V, 2A centre-tapped transformer (refer Fig. 2).

The transformer drives the bridge rectifier module BR1 and four 2200µF capacitors to provide +21V and -21V DC supplies. The left and the right channel input stages formed by the TL072 op-amp require $\pm 12V$ DC supplies for the operation. While +12V is obtained from resistor R12 and zener diode ZD1, -12V is obtained from resistor R23 and zener diode ZD2.